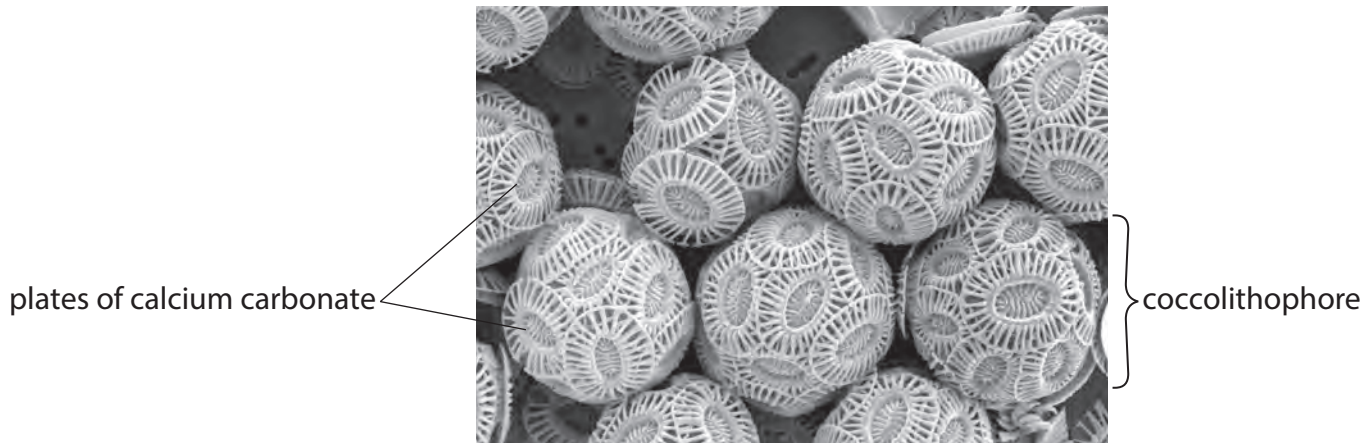


- 8 Phytoplankton are single-celled eukaryotic organisms that can photosynthesise. They live in the oceans.

A coccolithophore is a phytoplankton enclosed by plates of calcium carbonate.

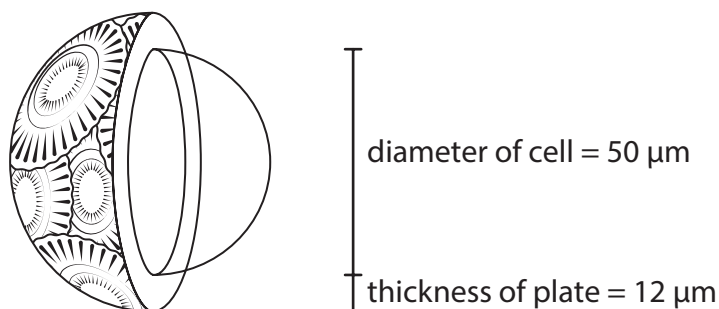
The photograph shows one species of coccolithophore, *E. huxleyi*.



(Source: © The Natural History Museum / Alamy Stock Photo)

- (a) In one drop of seawater there are between 1 000 and 10 000 *E. huxleyi*.

The diagram shows the dimensions of one *E. huxleyi*.



Calculate the **total** volume of this *E. huxleyi*.

Give your answer in mm^3 .

Use the formula:

$$V = \frac{4}{3}\pi r^3$$

(3)

Answer mm^3

- (b) Scientists think that *E. huxleyi* may have an impact on global warming and climate change.

E. huxleyi are continually producing the plates of calcium carbonate using the carbon dioxide they have taken in from the water.

- (i) One *E. huxleyi* can contain 12 plates.

It takes 75 minutes to produce one plate.

Calculate the number of **hours** it takes one *E. huxleyi* to produce 12 plates.

Assume only one plate at a time is made.

(1)

Answer hours

- (ii) Some of the plates produced by *E. huxleyi* are shed into the ocean water.

Scientists have discovered that plates shed into the water can attach to other *E. huxleyi*.

Describe how scientists could show that plates shed into the water can attach to **other** *E. huxleyi* cells.

(3)

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*(iii) The plates can carry viruses that infect the *E. huxleyi*.

Infected *E. huxleyi* produce a glue that sticks many cells and plates together to form a 'marine snow'.

The marine snow is very dense and sinks to the bottom of the oceans.

Explain the possible impact that infected *E. huxleyi* could have on global warming and the effects of climate change.

Use the information in the question and your own knowledge to support your answer.

(6)

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

(Total for Question 8 = 13 marks)